

3.2.4 Cell recognition and the immune system

Content	Opportunities for skills development
Each type of cell has specific molecules on its surface that identify it. These molecules include proteins and enable the immune system to identify: • pathogens • cells from other organisms of the same species • abnormal body cells • toxins. Definition of antigen. The effect of antigen variability on disease and disease prevention. Phagocytosis of pathogens. The subsequent destruction of ingested pathogens by lysozymes. The response of T lymphocytes to a foreign antigen (the cellular response). • The role of antigen-presenting cells in the cellular response. • The role of helper T cells (T_H cells) in stimulating cytotoxic T cells (T_C cells), B cells and phagocytes. The role of other T cells is not required. The response of B lymphocytes to a foreign antigen, clonal selection and the release of monoclonal antibodies (the humoral response). • Definition of antibody. • Antibody structure. • The formation of an antigen-antibody complex, leading to the destruction of the antigen, limited to agglutination and phagocytosis of bacterial cells. • The roles of plasma cells and of memory cells in producing primary and secondary immune responses. The use of vaccines to provide protection for individuals and populations against disease. The concept of herd immunity. The differences between active and passive immunity.	

Content	Opportunities for skills development
Structure of the human immunodeficiency virus (HIV) and its replication in helper T cells. How HIV causes the symptoms of AIDS. Why antibiotics are ineffective against viruses. The use of monoclonal antibodies in: - targeting medication to specific cell types by attaching a therapeutic drug to an antibody - medical diagnosis. Details of the commercial or scientific production of monoclonal antibodies are not required. Ethical issues associated with the use of vaccines and monoclonal antibodies. The use of antibodies in the ELISA test. Students should be able to: - discuss ethical issues associated with the use of vaccines and monoclonal antibodies - evaluate methodology, evidence and data relating to the use of vaccines and monoclonal antibodies.	